



SECJ2203: Software Engineering

Project Proposal

Project Title : **Enhanced Bug Tracker**

Date : **Friday, 19 April 2024, 11:59 PM**

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1. Introduction

In the fast-paced world of software development, effective bug tracking and issue management are essential for ensuring the quality and reliability of software products. However, traditional bug tracking systems often suffer from usability issues, lack of essential features, and inefficiencies in the management process. To address these challenges and enhance the bug tracking experience for software development teams, our project aims to develop an "Enhanced Bug Tracker" system.

1.1 The Goal

The primary goal of this project is to design and implement an enhanced bug tracking system that offers improved usability, functionality, and user experience. By enhancing the existing bug tracking system, we aim to streamline the process of identifying, reporting, prioritizing, and resolving issues, thereby facilitating smoother collaboration and communication among team members.

1.2 The Scope

The scope of this project encompasses the development of a user-friendly interface, implementation of additional features, optimization of system performance, and deployment on a scalable infrastructure. By leveraging modern technologies such as PHP, cloud hosting, and MariaDB, we intend to create a robust and reliable bug tracking solution that meets the needs of software development teams across various domains.

Objectives:

- i. Design a user-friendly interface: We want our bug tracker to be simple and intuitive, so anyone on the team can quickly learn how to use it.
- ii. Implement additional features: We'll add features like email notifications, attachment support, and advanced search capabilities to make bug tracking more efficient.
- iii. Optimize performance and reliability: Our bug tracker will be fast, reliable, and scalable, ensuring smooth operation even when lots of users are using it at the same time.
- iv. Deploy on a scalable infrastructure: We'll put our bug tracker on the internet, so team members can access it from anywhere, anytime, making collaboration easier than ever before.

2. Software Process Model

This section of the proposal aims to outline the ideal software process model for the development of our "Enhanced Bug Tracker" system. We'll provide an explanation of the goal of the software process model, justify our choice of model, illustrate the chosen model's phases, and describe detailed activities relevant to each phase.

2.1 The Goal : A software process model, also known as a software methodology, is like a roadmap that guides the development of software from start to finish. It's a series of interconnected actions that ensure tasks are performed efficiently and effectively to achieve the desired final product. The goal of a software process model is to provide direction and coordination so that the software meets its objectives in the most efficient way possible.

There are various process models available, each with its own set of tasks, inputs, outputs, preconditions, and postconditions. These models define the sequence and flow of activities involved in software development, helping teams stay organized and focused throughout the project lifecycle.

2.2 The Chosen Software Process Model : For our project, we have chosen the Agile Processes methodology as the ideal software process model. Agile is well-suited for our project because it emphasizes flexibility, adaptability, and collaboration, which are essential for developing an enhanced bug tracking system that meets the evolving needs of software development teams.

In Agile, the development process is divided into short iterations, typically lasting 1-4 weeks. Each iteration results in a potentially shippable product increment, allowing for frequent feedback and adjustments based on changing requirements. This iterative approach enables us to respond quickly to user feedback, incorporate new features, and deliver value to customers faster.

Agile also promotes communication and collaboration between stakeholders and developers, ensuring that everyone is aligned on project goals and priorities. It encourages creativity and innovation among team members, empowering them to experiment with different ideas and solutions to address complex problems.

2.3 Illustration of Chosen Software Process Model :



The Agile process model consists of several phases, including:

Requirement Analysis: Gathering and prioritizing user requirements.

Design: Creating a design for the bug tracking system based on the requirements.

Implementation: Coding and building the system according to the design.

Testing: Testing the system to ensure it meets quality standards and functions as intended.

Deployment: Deploying the system to production and making it available to users.

Maintenance: Providing ongoing support and updates to the system to address issues and incorporate feedback.

2.4 Detailed Activities for Each Phase : Requirement Analysis: Conduct user interviews, surveys, and workshops to gather requirements. Create user stories and prioritize them based on business value.

Design: Develop wireframes, mockups, and prototypes to visualize the system. Define the architecture, database schema, and user interface design.

Implementation: Write code according to the design specifications. Follow coding standards and best practices to ensure code quality and maintainability.

Testing: Perform unit tests, integration tests, and system tests to verify the functionality and quality of the system. Use automated testing tools to streamline the testing process.

Deployment: Deploy the system to a staging environment for user acceptance testing. Once approved, deploy the system to production servers.

Maintenance: Monitor the system for bugs, performance issues, and security vulnerabilities. Provide regular updates and enhancements based on user feedback and changing requirements.

2.5 Project Timeline and Deliverables :

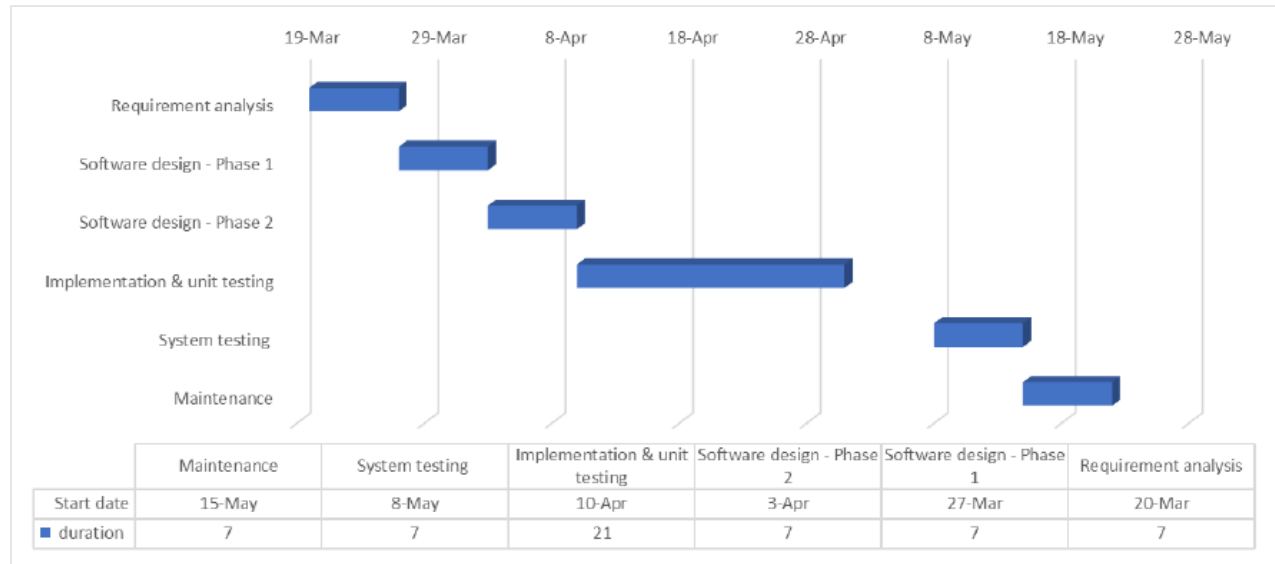


Figure 2.2 : Gantt Chart

- **Week 1-2:** Requirement Analysis (Deliverable: User Stories)
- **Week 3-4:** Design (Deliverable: System Design Document)
- **Week 5-6:** Implementation (Deliverable: Working Prototype)
- **Week 7-8:** Testing (Deliverable: Test Plan and Test Cases)
- **Week 9-10:** Deployment (Deliverable: Deployed System)
- **Week 11-12:** Maintenance (Deliverable: Maintenance Plan)